

Matematikos formulynai

10 klasė

Trigonometrinės tapatybės

- $\sin^2 \alpha + \cos^2 \alpha = 1$
- $\alpha = \frac{\sin \alpha}{\cos \alpha}, \alpha = \frac{\cos \alpha}{\sin \alpha}$
- $1 + \tan^2 \alpha = \frac{1}{\cos^2 \alpha}$

Suma ir skirtumas

- $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$
- $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$

Dvigubas kampas

- $\sin 2\alpha = 2 \sin \alpha \cos \alpha$
- $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 2 \cos^2 \alpha - 1$

Mažinimo formulės

- $\sin \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{2}}$
- $\cos \frac{\alpha}{2} = \pm \sqrt{\frac{1 + \cos \alpha}{2}}$